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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SBS-W BUS TIE H06																																																																								
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
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	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB3	HEATER LOCATED IN MOTOR CABINET							
	-BGK	POSITION SWITCH OF KEY LOCK						-EB4	HEATER FOR VENTILATION TEST							
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
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	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
D	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							D
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							
	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
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	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BM	HYGROSTAT						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
F	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							F
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFI	INTEGRATED CIRCUITS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFM	MICROPROCESSORS							
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFN	ANTIPUMPING RELAYS							
	-BU	BUCHHOLZ RELAY						-KFO	OPENING RELAYS OR CONTACTORS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFR	RECLOSING RELAYS							
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFS	SYNCHRONIZING DEVICES							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFZ	CONTROLLERS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							A		
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS									
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS									
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS									
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK									
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING									
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH									
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES									
	B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							B	
		-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES								
-PFB		BLUE SIGNAL LAMPS						-RAI	INDUCTORS									
-PFF		FLAG RELAYS						-RAR	RESISTORS									
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS									
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE									
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION									
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD									
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1									
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2									
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2									
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS									
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4									
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5									
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES									
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS									
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS									
	D	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS							D	
-PGJ		ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS									
-PGK		REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES									
-PGM		MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS									
	-PGP	POWER-FACTOR METERS						-SFT	TEST PUSH-BUTTONS									
	-PGQ	VARMETERS						-SFU	UNLOCKING PUSH-BUTTONS									
	-PGS	SYNCHRONOSCOPES						-SFV	VOLTMETRIC SWITCHES									
	-PGV	VOLTMETERS						-TA	POWER TRANSFORMERS									
	-PGW	WATTMETERS						-TB	CONVERTER									
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB1	RECTIFIER FOR FIRST OPENING RELEASE									
	-QAB	CIRCUIT-BREAKERS						-TB2	RECTIFIER FOR SECOND OPENING RELEASE									
	-QAC	CONTACTORS (FOR POWER)						-TB3	RECTIFIER FOR CLOSING RELEASE									
	-QBD	DISCONNECTORS						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE									
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER									
	-QBD4	TWO-WAY DISCONNECTOR														F		
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER																
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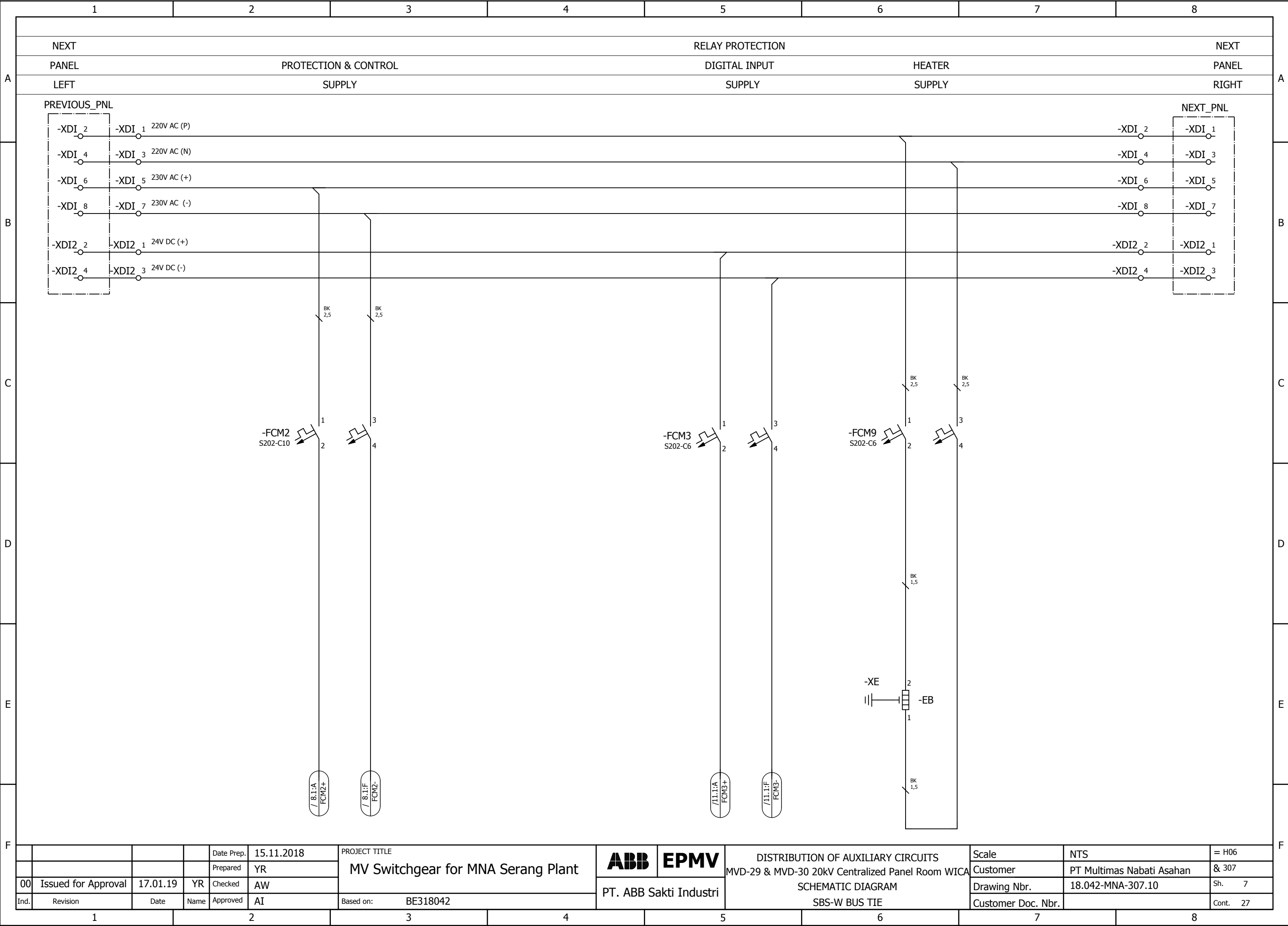
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A	-TFA	ACTIVE POWER TRANSDUCERS						-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS							A
	-TFC	CURRENT TRANSDUCERS						-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING							
	-TFF	FREQUENCY TRANSDUCERS						-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION							
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS							
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS							
B	-TFM	MULTIFUNCTION TRANSDUCERS						-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS							B
	-TFP	POWER-FACTOR TRANSDUCERS						-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS							
	-TFQ	REACTIVE POWER TRANSDUCERS						-XDI9	TERMINAL BLOCK FOR SPECIAL USE							
	-TFS	SIGNAL CONVERTERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD							
	-TFV	VOLTAGE TRANSDUCERS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS							
C	-WA	BUSBARS						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT							C
	-WBC	POWER CABLES						-XDS	CURRENT SOCKETS							
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB							
	-WGC	CONTROL CABLES						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
D	-WH	OPTICAL FIBRES						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS							D
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1							
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2							
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3							
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR							
E	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDX	SUPPORT TERMINAL BLOCK							E
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE							
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE							
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE							
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE							
F	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE							F
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE							
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT							
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR														
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET														
G	-XDC	CUSTOMER TERMINAL BLOCK														G
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER														
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR														
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT														
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS														
H	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER														H
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
I	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS														I
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR														
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS														
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS														
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)														
J	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)														J

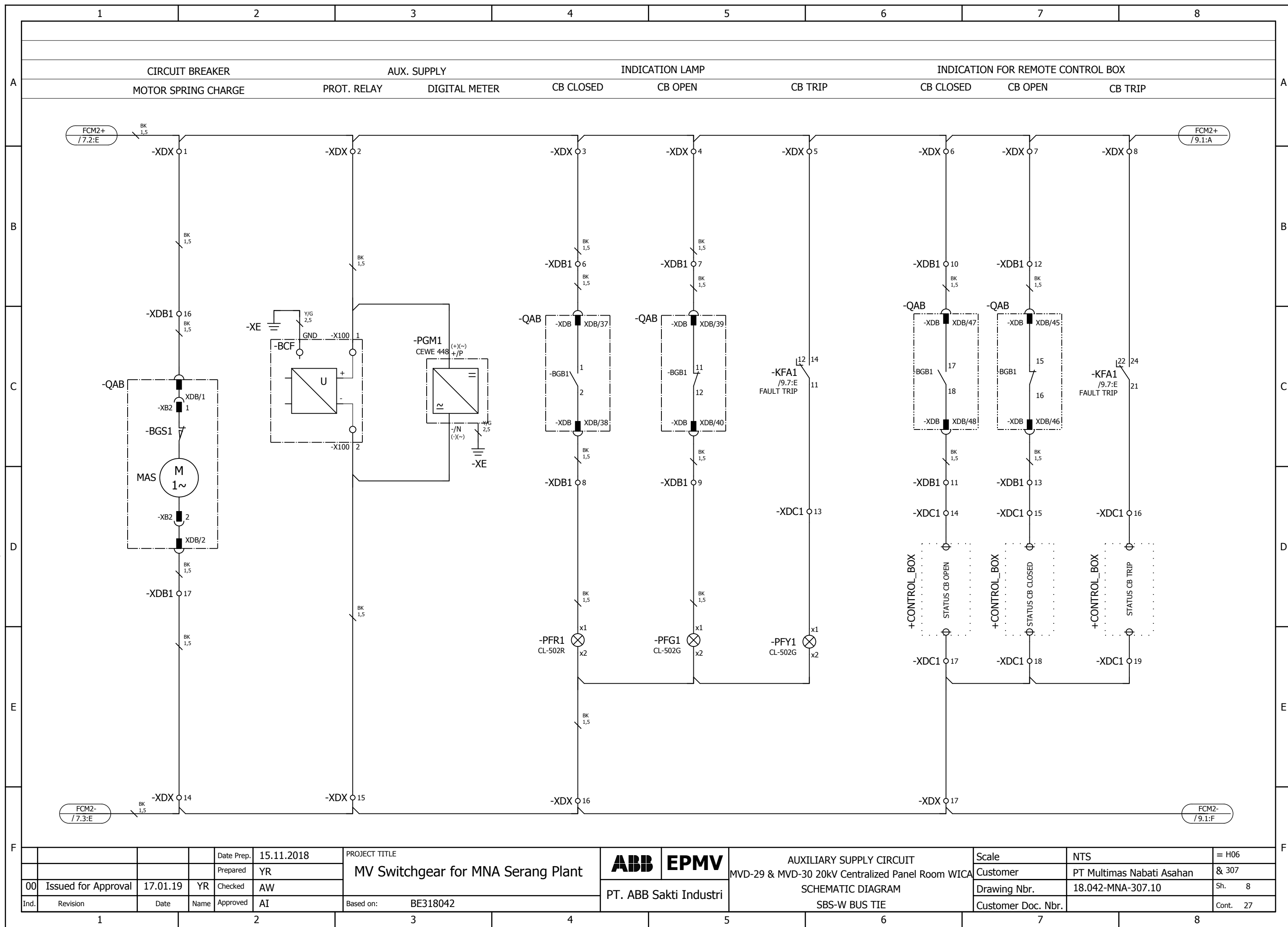
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant				REFERENCE DESIGNATIONS MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS		= H06
				Prepared	YR						Customer	PT Multimas Nabati Asahan		& 307
00	Issued for Approval	17.01.19	YR	Checked	AW						Drawing Nbr.	18.042-MNA-307.10		Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri		SBS-W BUS TIE	Customer Doc. Nbr.			Cont. 27

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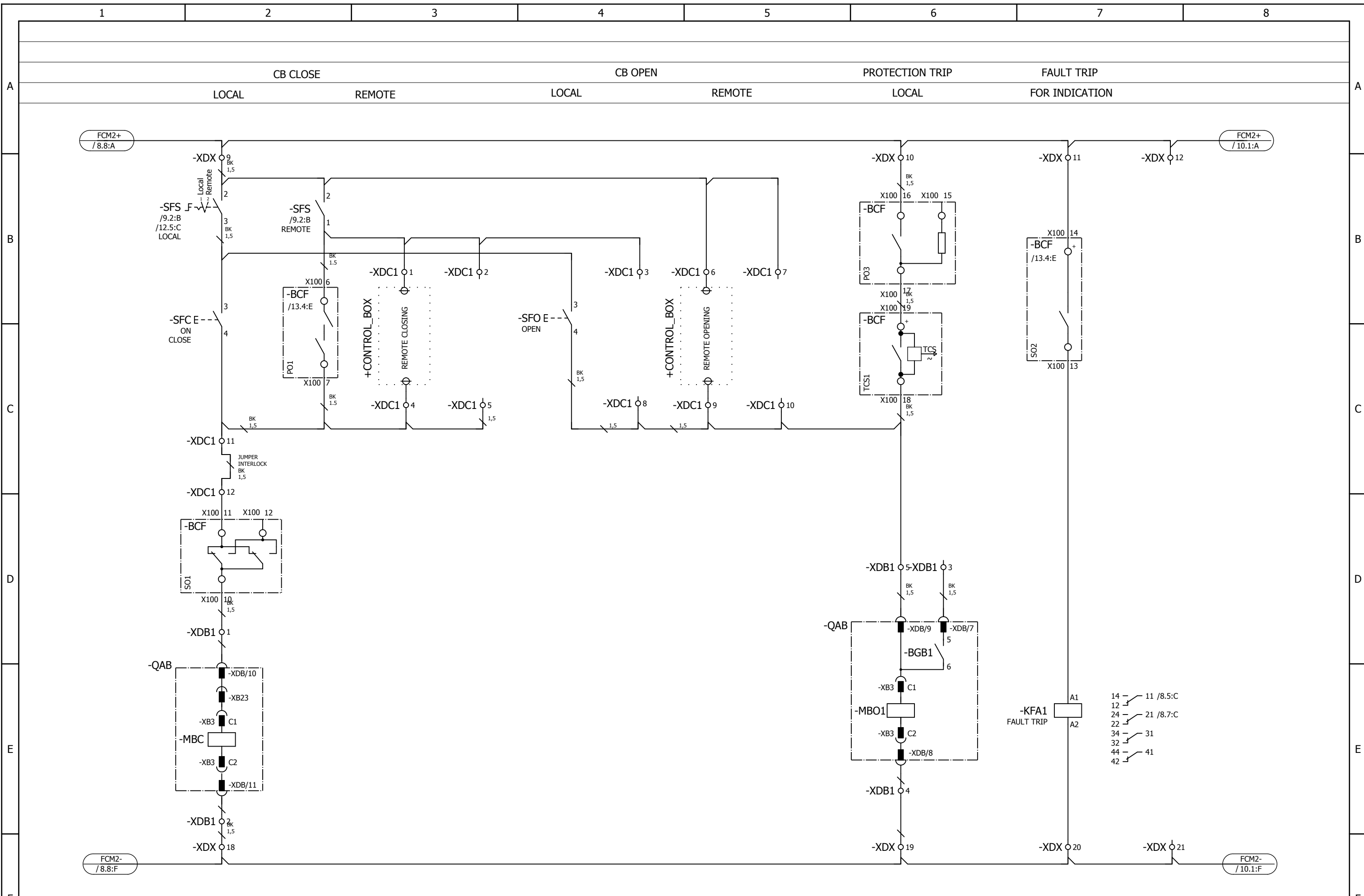
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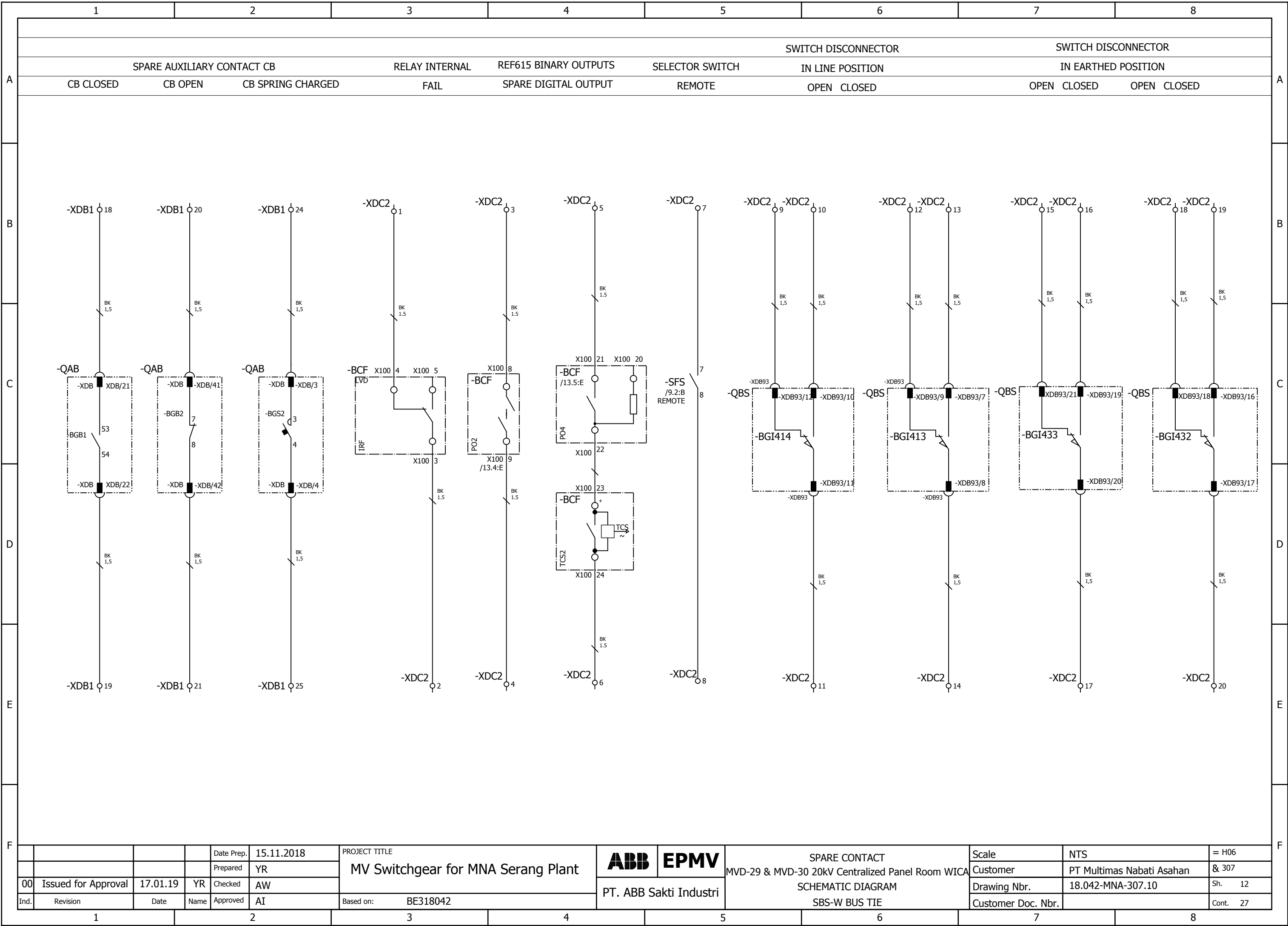


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	CIRCUIT BREAKER CONTROL MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H06
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 307
00	Issued for Approval	17.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-307.10	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBS-W BUS TIE	Customer Doc. Nbr.		Cont. 27

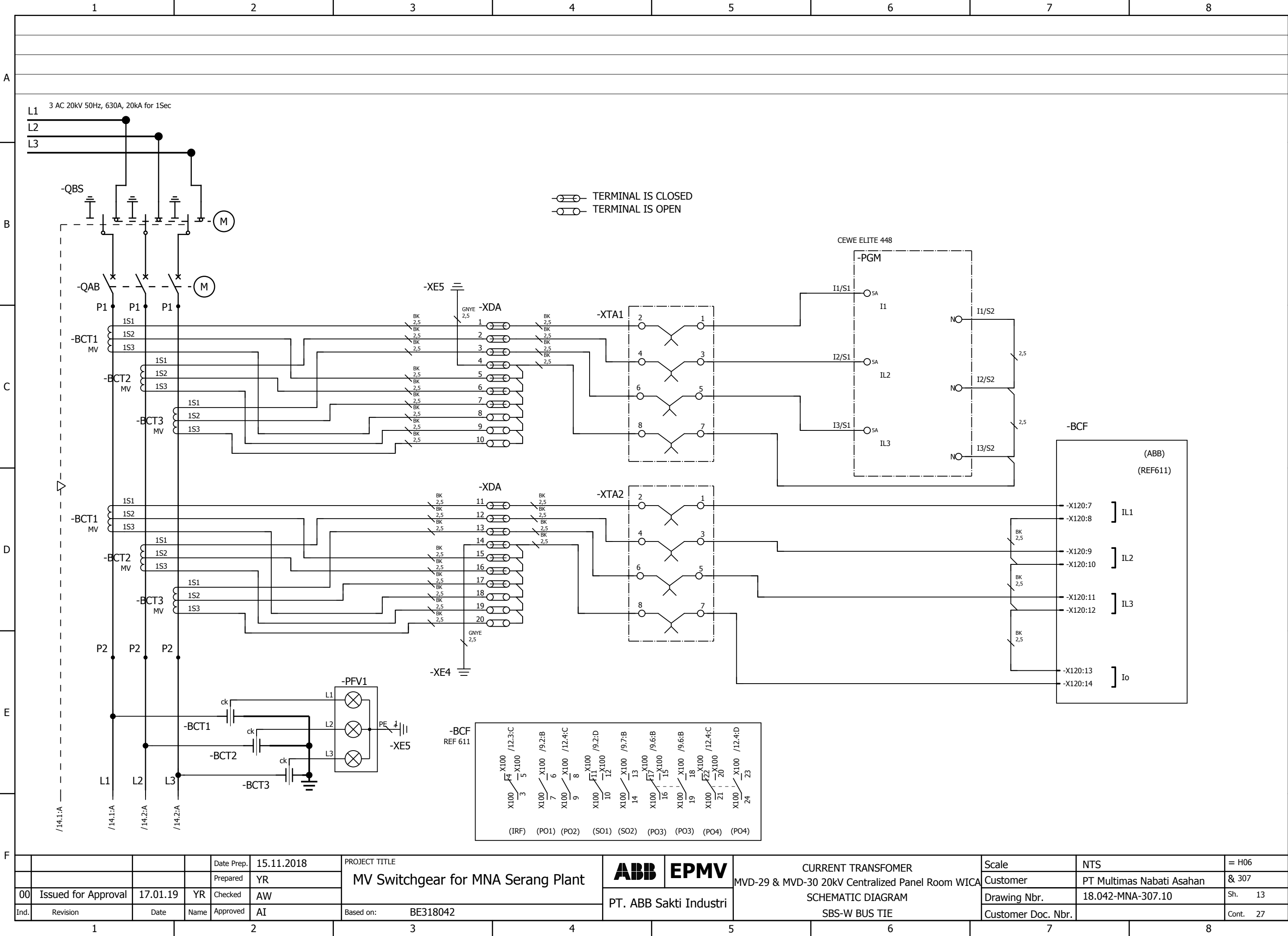
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H06
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 307
00	Issued for Approval	17.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-307.10	Sh. 11
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		SBS-W BUS TIE	Customer Doc. Nbr.		Cont. 27	

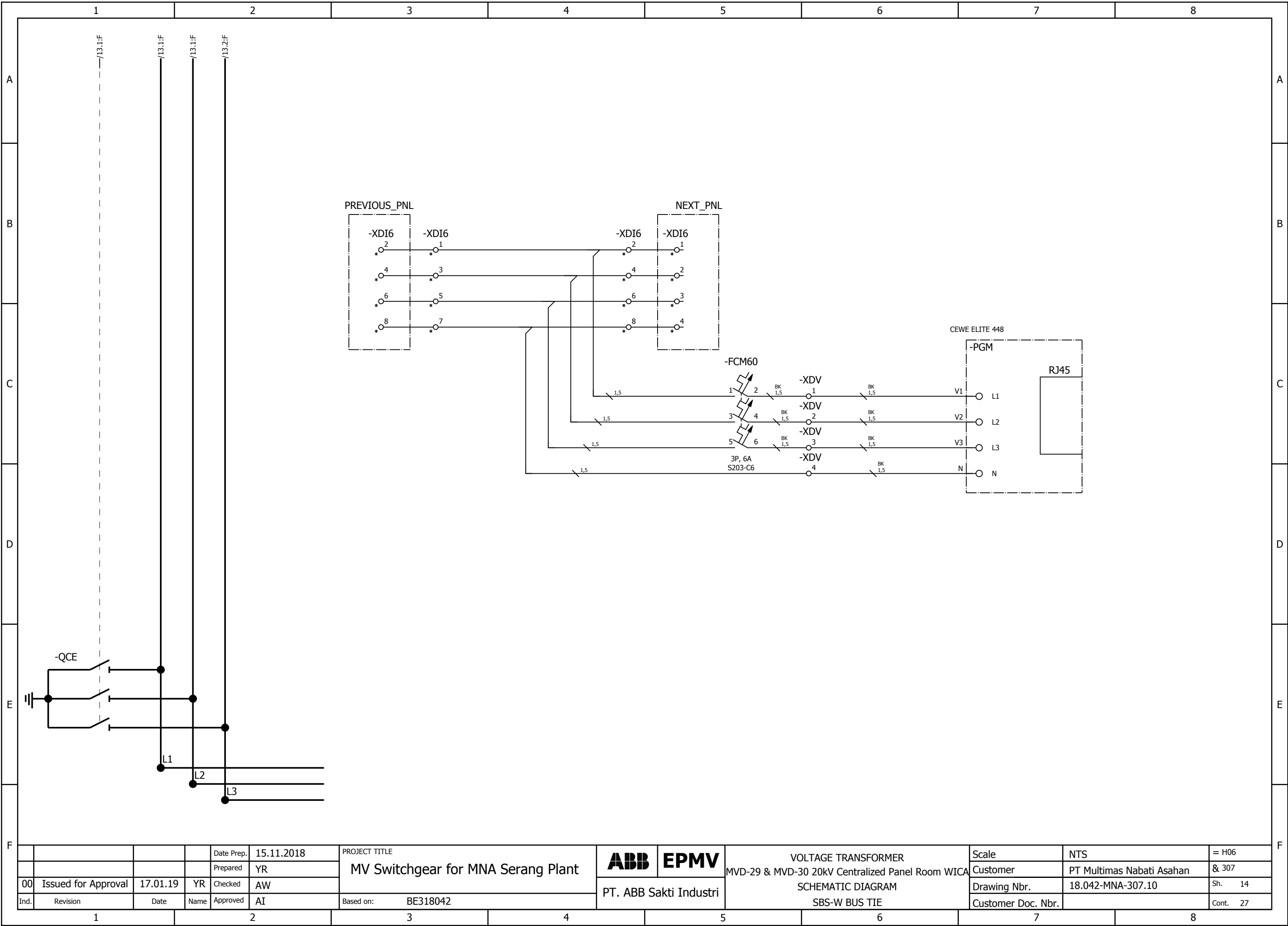
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			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	CURRENT TRANSFORMER	Scale	NTS	= H06	
			Prepared	YR			MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Customer	PT Multimas Nabati Asahan	& 307	
00	Issued for Approval	17.01.19	YR	Checked			AW	SCHMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-307.10	Sh. 13
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri	SBS-W BUS TIE	Customer Doc. Nbr.	Cont. 27



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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	VOLTAGE TRANSFORMER MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H06	
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 307	
00	Issued for Approval	17.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-307.10	Sh.	14
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	SBS-W BUS TIE			Customer Doc. Nbr.		Cont.	27

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

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A	Summarized parts list							IDABB_F02_005-IDABB					
								Carry over: 0.00					
	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS					
	-BCF	Feeder Management Relay REF611HBAABC1NN11G			ABB	REF611HBAABC1NN11G	1	/8.2:C					
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.6:E					
	-FCM2	MCB S202-C10 MCB. 2P, 10A			ABB	2CDS252001R0104	1	/7.2:C					
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	2	/7.5:C;/7.6:C					
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/14.5:C					
	-KFA1	Auxiliary Relay 230AC 4COS			PHOENIX	1YRA3DL266230001	1	/9.7:E					
	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G			ABB	1SFA619402R5232	1	/8.5:E					
B	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R			ABB	1SFA619402R5231	1	/8.4:E					
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y			ABB	1SFA619402R5233	1	/8.6:E					
	-PGM1	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC			CEWE	1YRA4EA0350001	1	/8.3:C					
	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR			IDEC	1YRA3DA14700001	1	/9.2:B					
	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR			IDEC	1YRA3DA14700002	1	/9.4:B					
C	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)			K&N	1YRA3DA17940011	1	/9.2:B					
	-XTA1;-XTA2	Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSCTT-04C	2	/13.4:C;/13.4:D					
D													
E													
F													
				Date Prep.	15.11.2018	PROJECT TITLE		PART LIST		Scale	NTS	= H06	
				Prepared	YR	MV Switchgear for MNA Serang Plant		ABB	EPMV	MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Customer	PT Multimas Nabati Asahan	& 307
00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-307.10	Sh. 17
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBS-W BUS TIE	Customer Doc. Nbr.		Cont. 27
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Ind. Revision Date Name Approved				REMARK		CABLE		EQUIPMENT		Wire No		Terminal number		TYPE		WIRE NO.		EQUIPMENT		CABLE		REMARK									
				BK 1,5 mm ²				-PGM:V1				1		PT 2.5				-FCM60:2				BK 1,5 mm ²									
				BK 1,5 mm ²				-PGM:V2				2		PT 2.5				-FCM60:4				BK 1,5 mm ²									
				BK 1,5 mm ²				-PGM:V3				3		PT 2.5				-FCM60:6				BK 1,5 mm ²									
				BK 1,5 mm ²				-PGM:N				4		PT 2.5				-XD16:8				1,5 mm ²									
==MVD-29_MVD-30=H06+-XDV																															
REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer'). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.																															
PROJECT TITLE				MV Switchgear for MNA Serang Plant				ABB		EPMV		Terminal diagram				Scale		= H06													
Based on:				BE318042				PT. ABB Sakti Industri						Customer		PT Multimas Nabati Asahan				& 307											
Date Prep.				15.11.2018										Drawing Nbr.						Sh. 26											
Prepared				YR										Customer Doc. Nbr.						Cont. 27											
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						CABLE	EQUIPMENT	WIRE No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK
				Date Prep.	15.11.2018	==MVD-29_MVD-30=H06+-XDX								
				Prepared	YR									
				Checked	AW									
Ind.	Revision	Date	Name	Approved	AI									
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